

Stock prices are random variables that many people try to guess.

-Pick stocks from the Turkish stock market (Borsa İstanbul), which are active stocks at least since 2021, two from each of these sectors: banking, energy, food (total $2 \times 3 = 6$). Get their adjusted daily stock prices since the beginning of 2021.

-Also get the USD/TRY exchange rate since 2021.

Write a report answering the following:

- 1-Find the correlation coefficient between the daily stock prices and the daily USD/TRY exchange rate.
- 2-Convert the stock prices to USD. Find the correlation coefficient between the daily stock prices.
- 3-Define new random variables as the daily price change. Find the correlation coefficient between the daily stock price changes and the daily USD/TRY exchange rate change.
- 4- Convert the price changes to USD. Find the correlation coefficient between the daily stock prices changes.
- 5-Do you think that we can assume the central limit theorem for any of these random variables? Why or why not? Check your reasoning with the data.

Expected change in correlated random variables can be approximated very simply linearly with the correlation coefficient. For example, the expected change in X can be approximated by using the change in Y as

$$E \left[\frac{\Delta X}{\sigma_X} \right] \approx \rho(X, Y) \frac{\Delta Y}{\sigma_Y}$$

- 6-Considering the most highly correlated stock prices in USD, try to estimate the 2021-now price change of one of them using the data of the other one.
- 7-Discuss your findings for all of the above.
- 8-Extra done analysis and discussion will be awarded additionally for up to 100% extra points.